

Project Orion Scenario and Classical Mistakes

Scenario: Project Orion

Project Orion was initiated by a global logistics company to build an AI-driven platform to optimize delivery routes in real time. The project was highly visible and led by a newly promoted manager, Ravi, who was technically skilled but had limited leadership experience.

To move faster, Ravi decided to skip formal documentation and design reviews, allowing developers to code freely. Initially, progress seemed quick, but soon performance issues appeared, features overlapped, and testing was delayed. The data science and development teams clashed because database structures changed frequently. Under pressure, Ravi blamed the hardware when performance failed during a demo, and the team added more servers instead of fixing the core software issues. Eventually, the system became unstable, and the project was suspended.

Question for Students

Read the scenario carefully and identify at least one **classical mistake** from each of the following categories: - People-related - Process-related - Product-related - Technology-related

For each, explain: 1. What the specific mistake was
2. Why it fits that category
3. How it could have been prevented

Model Answer Key (Beginner Level)

1. People-related Mistakes

Mistake:

Ravi made decisions without consulting his team and lacked leadership experience. There were also communication gaps between developers and data scientists.

Why it fits:

It involves teamwork, leadership, and communication problems.

How to prevent:

- Choose a manager with both technical and leadership skills.
 - Encourage open communication and teamwork.
 - Hold regular team meetings for alignment.
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2. Process-related Mistakes

Mistake:

Skipped documentation, design reviews, and testing processes to save time. No structured workflow or version control system.

Why it fits:

It shows weak project management and lack of proper process structure.

How to prevent:

- Follow a clear method (e.g., Agile or Scrum).
 - Maintain documentation and conduct regular reviews.
 - Use proper version control and consistent testing cycles.
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3. Product-related Mistakes

Mistake:

Overlapping features, poor performance, and unclear architecture design led to a low-quality product.

Why it fits:

These affect the quality and structure of the final product.

How to prevent:

- Plan a solid architecture before development.
 - Conduct regular quality and performance reviews.
 - Integrate and test features properly.
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4. Technology-related Mistakes

Mistake:

Team blamed hardware instead of fixing inefficient software design; incompatible frameworks caused issues.

Why it fits:

These come from poor technology choices and incorrect technical assumptions.

How to prevent:

- Evaluate technology early in the project.
 - Use compatible, scalable tools and frameworks.
 - Fix root causes in software instead of adding hardware.
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Summary Table

Category	Example Mistake	Why It Fits	Prevention
People	Poor leadership & communication	Human/team issue	Train manager, improve collaboration
Process	Skipped documentation/ reviews	Weak process	Follow project methods
Product	Messy design & low quality	Product defect	Plan architecture, test early
Technology	Wrong diagnosis & poor tools	Tech inefficiency	Choose right tools, test scalability